

Nathan Soldati

Herbstweg 94, 8050 Zürich, CH

+41 79 933 22 52

nathan.soldati01@bluewin.ch

nsoldati.github.com



EDUCATION

ETH Zürich - MSc Information Technology and Electrical Engineering *2024 - Present*

Focus:

- Machine Learning, AI and Computer Vision
- Control Systems, Automation and Optimization

University of Melbourne *Spring 2025*

- Exchange Semester

ETH Zürich - BSc Information Technology and Electrical Engineering *2021 - 2024*

- **GPA:** 5.88/6.00 (top 2%)
- **Bachelor Thesis:** Multi-Layer Feedback Control for Selective Laser Melting Using a Backpropagation Strategy.

ACADEMIC EXPERIENCE AT ETHZ

Teaching Assistant *2022 - 2024*

- **Courses:** Engineering Mechanics, Signal and Systems I and Signal and Systems II.
- **(AMIV) Exam Preparation Courses:** Selected twice (based on student surveys) to lead exam preparation sessions for approximately 120 electrical engineering students.

Research Assistant - Automatic Control Laboratory *Spring 2023*

- Preparation of theory scripts.
- Direct collaboration with professors and PhDs.

PROFESSIONAL EXPERIENCE

AI & Computer Vision Engineer at Swiss Engineering Partners AG *Feb 2025 - Present*

- Full-Time Internship
- Developing and deploying computer vision systems for automated industrial video analysis.
- Building scalable data pipelines and active learning workflows for continuous model improvement.
- Managing the full ML lifecycle from experimentation to production deployment.
- Collaborating with stakeholders to translate operational requirements into AI solutions.

- Part-Time
- Fine-tuned Vision Transformers (ViT) for semantic segmentation of architectural floor plans.

TECHNICAL SKILLS

Programming

- Python
- MATLAB/Simulink
- C/C++
- Assembly

Theory

- Deep theoretical understanding of ML models and algorithms.
- Model training, evaluation, and data-centric workflows.
- Optimization and numerical methods for engineering and AI applications.

Development Tools

- Linux, Git, Docker
- MLOps experiment tracking (MLflow, Weights & Biases, Tensorboard)
- Model deployment and APIs (FastAPI, REST services)
- Data processing frameworks (Pandas, NumPy)

Other Tools

- Latex, CAD, Office 365

AWARDS

- **Nizzola Excellence Scholarship:** Awarded for exceptional academic performance during bachelor's studies.
- **ITET Best BSc Thesis Award:** Awarded to the top 3 bachelor's students in the Electrical Engineering and Information Technology department at ETH Zürich.

LANGUAGES

- **Italian:** native speaker
- **English:** full professional knowledge
- **German:** professional knowledge
- **French:** good knowledge

REFERENCES

Available on request